

HARSH BHANUSHALI

AI Engineer

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SUMMARY

AI Engineer building multi-agent systems, RAG pipelines, and cost-aware LLM workflows in Python. Architecture-first approach focused on explicit state, deterministic routing, bounded execution, tool orchestration, and reliable system behavior.

SKILLS

AI / ML: Multi-Agent Systems, RAG, LLM Integration, Tool-Calling, Prompt Engineering, Function Calling, Embeddings, Vector Databases, Groq, Gemini
Languages & Libraries: Python, SQL, NumPy, Pandas, Pydantic, LangChain, LangGraph, SentenceTransformer
Frameworks & Tools: FastAPI, Streamlit, ChromaDB, MCP (Model Context Protocol)

EXPERIENCE

Symbiosis Systems and Services Pvt. Ltd.

Jan 2024 - Sep 2024

Full Stack Developer Intern

- Engineered and delivered 20+ features across 15+ backend modules in 3 enterprise web applications using ASP.NET and SQL Server, reducing manual workflow steps by 30% for daily active users.
- Architected and led the overhaul of the core enterprise application, reengineering the codebase to build automated cross-module data workflows that eliminated system friction.

PROJECTS

Waypoint - Multi Agent AI Trip Planner

Aug 2026 - Present

- Architected and implemented a 5-agent LangGraph workflow (Concierge -> Planner -> Budget -> Itinerary -> Critic) with a human-review gate for iterative itinerary refinement.
- Engineered an explicit TypedDict TripState as the single source of truth and integrated Postgres checkpointing for persistent graph state across refreshes and restarts.
- Technologies: Python · LangGraph · LangChain · FastAPI · PostgreSQL · Pydantic · Groq · MCP

AI Agent Engine

Feb 2026 - Present

- Architected a 4-layer routing pipeline (cache -> pattern matcher -> RAG -> LLM) that resolves ~80% of repeat queries without LLM inference, keeping inference cost near zero for common usage patterns.
- Implemented bounded execution with Planner -> Validator -> Executor -> Replanner -> Responder, including automatic retry/replanning, session-level tracing, and quota enforcement; local semantic routing runs at ~30 ms with an estimated cost of ~0.0005/session.
- Technologies: Python, LangChain, ChromaDB, Gemini, FastAPI, Pydantic

hArI — Document Intelligence App

Mar 2026 - Present

- Engineered hybrid retrieval combining pgvector semantic similarity and PostgreSQL tsvector full-text search through a custom Supabase RPC; replaced pandas.exec() with in-memory DuckDB for secure multi-CSV SQL queries.
- Integrated Supabase Auth, per-user chat persistence, and citation-aware responses showing the exact retrieved chunks used for each answer.
- Technologies: Python · Streamlit · Supabase · PostgreSQL · pgvector · DuckDB · Groq · LangChain

Multi-Agent Research Pipeline

Apr 2026 - Present

- Architected a 6-agent supervisor workflow (Supervisor -> Search -> Scraper -> Summarizer -> Critic -> Synthesizer) for deep, multi-hop web research.
- Engineered automatic gap detection that triggers re-search before synthesis; produces reports covering 30-50 sources across 5-9 targeted questions per run with complexity-based routing from Groq primary to Gemini fallback.
- Technologies: Python, Groq, Gemini, DuckDuckGo API, Rich CLI

FinOS - AI Finance Platform

May 2026 - Present

- Developed a full-stack finance platform combining an analytics dashboard and conversational agent over a shared SQLite store; pattern matching resolves ~60% of queries with 0 LLM calls.
- Added financial health scoring, spending forecasts, budget alerts, and bcrypt-hardened auth.
- Technologies: Python, FastAPI, SQLAlchemy, Groq, SSE, SQLite

EDUCATION

Government Engineering College Modasa, Modasa
B.E. in Information Technology

Sep 2020 - May 2024
CGPA: 8.23 | GATE 2025 CSE — Qualified